

Ron Quah

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Education

National University of Singapore

August 2023 – May 2027

Bachelor of Computer Science

Experience

Software Engineer Intern | *Python, RPC, Agent Development Kit*

May 2026 – August 2026

Google

Singapore

- Architected and implemented a **multi-agent orchestration platform** using **Agent Development Kit** to automate documentation maintenance across large-scale codebases, reducing engineer effort by **60%** by replacing manual remediation with agent-generated **pull requests** requiring only human review and approval
- Extended **Agent Development Kit** with custom **agent tool wrappers** over internal services, exposing standardised interfaces for secure agent interactions and making **reusable agent tooling available to 80+ engineering teams**
- Designed a robust **agent evaluation framework** using **autoraters and golden datasets**, combining **deterministic trajectory matching** to validate safe tool execution with **LLM-as-a-judge** evaluation to measure semantic correctness and reduce hallucinations

Backend Engineer Intern, Data Platform | *Spring, Kafka, Redis, Grafana* December 2025 – April 2026

Shopee Singapore

- Designed and implemented a **distributed log storage pipeline** using **Kafka** and **ClickHouse**, mapping **Kafka partitions to ClickHouse shards**, configuring **ReplicatedMergeTree replicas** for fault tolerance, and defining **partition and sort keys** for efficient time-range log queries, supporting log ingestion of **1M+ task instances daily**
- Improved **data quality service reliability** by instrumenting **Grafana monitoring** for JVM and workload metrics and tuning **thread pool size, queue capacity, and concurrency limits**, reducing memory usage by **30%** and preventing resource contention under peak load
- Built a **centralised self-service dashboard** for configuring compute resources for **Spark and Presto data quality jobs**, strengthening **data governance** through **ownership-based access control** that restricted checks to authorised database owners while exposing workload settings such as **CPU, memory, and concurrency limits**

Software Engineer Intern, Global KYC | *Spring, MySQL, Kafka, TestNG* May 2025 – November 2025

Binance Singapore

- Designed and implemented a scalable **Spring Boot microservice** for a multi-stage document verification pipeline, orchestrating OCR extraction and **LLM-based compliance checks** with **RAG-driven jurisdiction rules**, generating explainable pass/fail previews and reducing manual compliance review time by **40%**.
- Engineered a fault-tolerant **Kafka** fallback mechanism using **Redis** as a temporary persistence layer during broker unavailability and **ZooKeeper** leader re-election, enabling reliable message recovery and reprocessing after cluster stabilisation.

Junior Developer | *React, Express.js, Tailwind, JavaScript*

November 2024 – May 2025

Reform DAO

Remote

- Built full-stack airdrop features by integrating **React** UIs with **Express.js** APIs, designing client-server data flows to support up to **3,000 users** during peak usage.
- Architected data-driven dashboards using **TanStack Query** for API orchestration and caching, and **TanStack Table** for efficient client-side data presentation, improving responsiveness and scalability of frontend data handling.

FrontEnd Developer Intern | *React, Next.js, Tailwind, JavaScript*

July 2024 – September 2024

Classpoint

Singapore

- Architected and maintained responsive web interfaces using **React** and **Next.js**, designing reusable component hierarchies and Tailwind-based design systems to enable consistent UI composition and rapid iteration.
- Collaborated cross-functionally to deliver **10+ production pages**, aligning frontend architecture with product requirements and backend APIs, and ensuring clear separation between presentation, state management, and data-fetching concerns.
- Optimised client-side performance by refactoring component trees and leveraging **Next.js dynamic imports**, lazy loading, and code-splitting strategies to reduce initial bundle size and improve page load times.

Technical Skills

Languages: Python, JavaScript, Java, Golang, TypeScript, HTML/CSS, SQL (PostgreSQL, MySQL), MongoDB, ClickHouse

Developer Tools: VS Code, Docker, GitHub, Jest

Frameworks: React, Next.js, Spring Boot, Express.js, TensorFlow, PyTorch, Spark

Technologies: Apache Kafka, Apache Spark, Redis, Firebase, ClickHouse, Grafana, Google Cloud Provider (GCP), AWS (Amazon Web Services)